Jiaxuan Lin

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Research Interests

My research interests lie in robot learning for dexterous manipulation. I am particularly interested in learning from human videos and physical interaction, with reinforcement learning and vision-language-action (VLA) models.

Education & Research Positions

University of Science and Technology of China (USTC) Sep 2024 – Jun 2028 (expected)
B.Tech. in Artificial Intelligence • GPA 3.86/4.30 • Rank 4/59 Hefei, China

- **Selected coursework:** Foundations of AI and Machine Learning (99), Data Structures (96), Mathematical Analysis (93)

National University of Singapore (NUS) Mar 2026 – Present
Visiting Research Intern • Advisor: Prof. David Hsu Singapore

Publications

[P1] CAC-VLA: Context-Gated Action Conditioning for Vision-Language-Action Models
Yifu Xiong, Wenhao Yu, **Jiaxuan Lin**, Bojun Zou, Jiahao Li, Lu Zhang, Yanyong Zhang, Jianmin Ji
Under review [arXiv]

[P2] VLA-Stem: A Lightweight Safety Mechanism for Vision-Language-Action Models via Reinforcement Learning
Bojun Zou, Wenhao Yu, Jiahao Li, **Jiaxuan Lin**, Guanting Ye, Haohan Geng, Weihao Yu, Yanyong Zhang, Jianmin Ji
Under review [arXiv]

Research Experience

Feasibility-Guided Dexterous Manipulation Jul 2026 – Present
Project Lead • NUS • Advisor: Prof. David Hsu

- Goal-conditioned RL without reference trajectories takes a 22-DoF Sharpa Wave hand from an object on a table to a fully specified grasp (palm pose, joints, contacts), re-posing the object first when the table blocks the grasp.
- Adapted Dexonomy to the hand (6.4M grasps) and built an RL search for table-feasible grasps near these references.
- Staged PPO (SAPG + LSTM) with mechanism-level fixes raised success from 77.7% to 98.6% (16/16 targets, sim).
- Traced a training collapse to a critic-blind null space in MIT-mode actions and removed it with an analytic torque split; gain alignment keeps 97.7% success under firmware gain limits (67% at defaults).

Continual Dexterous Skill Learning for VLAs from Human Video Jul 2026 – Present
Project Lead • USTC • Advisor: Prof. Yanyong Zhang

- Grow skills without forgetting via MoE expert re-routing, lightweight new capacity, and periodic distillation.
- Condition expert routing on egocentric human videos (V-JEPA) as task-level guidance rather than exact trajectories.
- Repair retargeted human motion in physics-validated simulation via LLM-agent DAgger for a dual-arm dexterous robot.

VLA-Stem: RL-Based Safety for VLA Models [P2] [Code] Feb 2026 – May 2026
USTC • Advisor: Prof. Jianmin Ji

- Implemented AprilTag-based 6-DoF tracking of moving obstacles for real-robot safety evaluation.
- Extended a Quest 3 XR interface into a ROS 2 teleoperation and data-collection system for a UR7e (up to 90 Hz, RLDS).

Robot Systems & Engineering

AutoLife Simulation & Whole-Body Control Platform [Code] Mar 2026 – Jun 2026
NUS • Advisor: Prof. David Hsu

- Built AutoLife in Isaac Sim with RGB-D/LiDAR/IMU sensing and MolmoSpaces and BEHAVIOR-1K scenes.
- Implemented a ROS 2 whole-body control stack integrated with the AutoLife planner for pre-deployment validation.

Occlusion-Robust 3D Manipulation with SAM 3D Dec 2025 – Mar 2026
USTC • Advisor: Prof. Jianmin Ji

- Built a calibrated RGB-D pipeline that completes occluded objects with SAM 3D for AnyGrasp/AnyPlace pick-and-place.

Competition

RoboCup@Home 2026 World Finals, Incheon • **Team WrightEagle.AI (USTC)** [Project Page] Jul 2025 – Jul 2026
Core Member, owner of the Pick and Place Challenge (kitchen manipulation) • Advisor: Prof. Jianmin Ji

- Post-trained $\pi_{0.5}$ (full-parameter, 8×A100) on the Galaxea Open-World Dataset for the R1 Lite bimanual robot.
- Learned atomic bimanual skills for breakfast preparation (cap unscrewing, pouring, table setup) from teleoperation.
- Integrated zero-shot 6-DoF grasping (Grounded-SAM 2, GraspNet, MoveIt 2) and remote GPU inference over ROS 2.

Honors & Awards

First Prize, Artificial Intelligence Honors Program Scholarship	2025, 2026
Gold Award, Outstanding Student Scholarship (1 awarded per year in the major)	2025
Silver Award, Outstanding Student Scholarship (5 awarded per year in the major)	2026
Silver Award, Outstanding Freshman Scholarship	2024

Technical Skills

Robot platforms	Galaxea R1 Lite (bimanual mobile manipulator), UR7e, RealMan arms
Robotics & simulation	ROS / ROS 2, MoveIt 2, NVIDIA Isaac Sim, Isaac Lab, MuJoCo
Learning & programming	PyTorch, JAX, Python, C/C++